# HDOC-36

## Aim:

To write a menu-driven C program using separate functions to perform various string operations:

1. Count the number of characters in a string without using built-in length functions.
2. Print each character of the string on a new line.
3. Count the number of vowels and consonants in the string.
4. Exit the program.

## Task 1: Algorithm:

1. Start the program.
2. Declare a character array str[100] and an integer choice.
3. Accept a string from the user using fgets().
4. Display the menu repeatedly until the user selects Exit.
5. According to the user's choice:
    - Choice 1: Call countCharacters() and count characters until '\0' or '\n' is reached.
    - Choice 2: Call printCharacters() and print every character on a separate line.
    - Choice 3: Call countVowelsConsonants(). Check each alphabetic character and classify it as a vowel or consonant.
    - Choice 4: Exit the program.
6. For any other choice, display Invalid choice.
7. Stop.

## Task 2: Test Case Table:

| Test Case | Input String | Choice | Operation | Expected Output |
|---|---|---|---|---|
| 1 | Hello | 1 | Count characters | Number of characters = **5** |
| 2 | CAT | 2 | Print each character | C, A, T printed on separate lines |
| 3 | Programming | 3 | Count vowels & consonants | Vowels = **3**, Consonants = **8** |

## Task 3: C program:

```
#include <stdio.h>
int countCharacters(char str[]);
void printCharacters(char str[]);
void countVowelsConsonants(char str[]);
int main()
{
   char str[100];
   int choice;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);
   do
   {
      printf("\n----- MENU -----\n");
      printf("1. Count characters in the string\n");
      printf("2. Print each character on a new line\n");
      printf("3. Count vowels and consonants\n");
      printf("4. Exit\n");
      printf("Enter your choice: ");
      scanf("%d", &choice);
      switch (choice)
      {
         case 1:
            printf("Number of characters = %d\n",
                  countCharacters(str));
            break;
```

```
            case 2:
                printf("Characters in the string:\n");
                printCharacters(str);
                break;
            case 3:
                countVowelsConsonants(str);
                break;
            case 4:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice! Please enter 1 to 4.\n");
        }
    } while (choice != 4);
    return 0;
}
/* Function to count characters without strlen() */
int countCharacters(char str[])
{
    int count = 0;
    while (str[count] != '\0' && str[count] != '\n')
    {
        count++;
    }
    return count;
}
/* Function to print each character on a new line */
void printCharacters(char str[])
{
    int i = 0;
    while (str[i] != '\0' && str[i] != '\n')
    {
        printf("%c\n", str[i]);
        i++;
    }
}
/* Function to count vowels and consonants */
void countVowelsConsonants(char str[])
{
    int i = 0;
    int vowels = 0;
    int consonants = 0;
    char ch;
    while (str[i] != '\0' && str[i] != '\n')
    {
        ch = str[i];
```

```
      if ((ch >= 'A' && ch <= 'Z') ||
         (ch >= 'a' && ch <= 'z'))
      {
         if (ch == 'A' || ch == 'E' || ch == 'I' ||
            ch == 'O' || ch == 'U' || ch == 'a' ||
            ch == 'e' || ch == 'i' || ch == 'o' ||
            ch == 'u')
         {
            vowels++;
         }
         else
         {
            consonants++;
         }
      }
      i++;
   }
   printf("Number of vowels = %d\n", vowels);
   printf("Number of consonants = %d\n", consonants);
}
```

# Task 4: output:

